

Differential Equations (2022–2026)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

44

TOTAL MARKS

71 M

TEST DURATION

132 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	23	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	15	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q51 • ID: JAM_2026_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51

Let α, β and γ be fixed real numbers such that $y(x) = C_1 e^{-x} + C_2 e^{2x} + \alpha x e^{-x}$, where C_1 and C_2 are arbitrary real constants,

is the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + \beta \frac{dy}{dx} + \gamma y = -e^{-x}.$$

Then $\alpha(\beta + \gamma) =$ _____ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q45 • ID: JAM_2026_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45

Let α and β be real numbers such that the differential equation

$$(y^3 + \alpha xy^4 - 5x + \cos 2y)dx + (3xy^2 + 20x^2 y^3 + \beta x \sin 2y)dy = 0$$

is exact.

Then $\alpha + \beta =$ _____ (rounded off to one decimal place).

Question 3 (JAM 2026 • Q35 • ID: JAM_2026_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35	Consider the differential equation $(2y \cos x - xy \sin x) dx + 2x \cos x dy = 0 \quad \text{for } x \in \left(0, \frac{\pi}{4}\right).$ Which of the following is/are integrating factor(s) of the differential equation?
(A)	$\frac{1}{xy}$
(B)	xy
(C)	$\sec x$
(D)	$\sqrt{\sec x}$

Question 4 (JAM 2026 • Q23 • ID: JAM_2026_Q23)

MCQ

+2 M

Neg: -0.67

Q.23	Consider the differential equation $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y = \sin(e^x).$ Which ONE of the following is a particular solution of this differential equation?
(A)	$-e^{-x} \sin(e^x)$
(B)	$-e^{-2x} \sin(e^x)$
(C)	$-e^{-x} (\sin(e^x) + \cos(e^x))$
(D)	$-e^{-2x} (\sin(e^x) + \cos(e^x))$

Question 5 (JAM 2026 • Q18 • ID: JAM_2026_Q18)

MCQ

+2 M

Neg: -0.67

Q.18	Let $\phi: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be the solution of the differential equation $(\cos x) \frac{dy}{dx} - y = 2y^2(\sin x - 1) \cos x$ satisfying $\phi(0) = \frac{1}{2}$. Then $\phi\left(\frac{\pi}{4}\right) =$_____.
(A)	2
(B)	$\frac{3 + 2\sqrt{2}}{2}$
(C)	$\frac{1}{2}$
(D)	$\frac{1}{\sqrt{2}}$

Question 6 (JAM 2026 • Q16 • ID: JAM_2026_Q16)

MCQ

+2 M

Neg: -0.67

Q.16	The orthogonal trajectories of the family of curves $y = -3x - 3 + me^x \quad \text{for a real parameter } m$ are given by _____.
(A)	$x = \frac{y}{3} - \frac{1}{9} + k e^{-3y}$, k is a real parameter
(B)	$x = \frac{y}{3} - \frac{1}{9} + k e^{3y}$, k is a real parameter
(C)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{-3y}$, k is a real parameter
(D)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{3y}$, k is a real parameter

Question 7 (JAM 2026 • Q4 • ID: JAM_2026_Q4)

MCQ

+1 M

Neg: -0.33

Q.4	The general solution of the differential equation $x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + \frac{x}{2}, \quad x > 0$ is given by _____.
(A)	$\cos\left(\frac{x}{y}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(B)	$\cos\left(\frac{y}{x}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(C)	$\cos\left(\frac{x}{y}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant
(D)	$\cos\left(\frac{y}{x}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant

Question 8 (JAM 2025 • Q56 • ID: JAM_2025_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} + 2xy = 2 + 4x^2$$

satisfying $\varphi(0) = 0$.

Then, the value of $\varphi(2)$ is equal to _____ (rounded off to two decimal places).

Question 9 (JAM 2025 • Q55 • ID: JAM_2025_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 Let $\varphi: (0, \infty) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 6x \ln x$$

satisfying $\varphi(1) = -3$ and $\varphi(e) = 0$.

Then, the value of $|\varphi'(1)|$ is equal to _____ (rounded off to two decimal places).

Question 10 (JAM 2025 • Q44 • ID: JAM_2025_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the differential equation

$$4 \frac{d^2 y}{dx^2} + 16 \frac{dy}{dx} + 25y = 0$$

satisfying $\varphi(0) = 1$ and $\varphi'(0) = -\frac{1}{2}$.

Then, the value of $\lim_{x \rightarrow \pi/6} e^{2x} \varphi(x)$ is equal to _____ (rounded off to two decimal places).

Question 11 (JAM 2025 • Q33 • ID: JAM_2025_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $\varphi : (-1, \infty) \rightarrow (0, \infty)$ be the solution of the differential equation

$$\frac{dy}{dx} - 2ye^x = 2e^x\sqrt{y},$$

satisfying $\varphi(0) = 1$.

Then, which of the following is/are TRUE?

(A) φ is an unbounded function

(B) $\lim_{x \rightarrow \ln 2} \varphi(x) = (2e - 1)^2$

(C) $\lim_{x \rightarrow \ln 2} \varphi(x) = \sqrt{2e - 1}$

(D) φ is a strictly increasing function on the interval $(0, \infty)$

Question 12 (JAM 2025 • Q32 • ID: JAM_2025_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 If $M, N, \mu, w: \mathbb{R}^2 \rightarrow \mathbb{R}$ are differentiable functions with continuous partial derivatives, satisfying

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = dw$$

then which of the following is/are TRUE?

- (A) μw is an integrating factor for $M(x, y)dx + N(x, y)dy = 0$
- (B) μw^2 is an integrating factor for $M(x, y)dx + N(x, y)dy = 0$
- (C) $w(x, y) = w(0, 0) + \int_0^x (\mu M)(s, 0)ds + \int_0^y (\mu N)(x, t)dt$, for all $(x, y) \in \mathbb{R}^2$
- (D) $w(x, y) = w(0, 0) + \int_0^x (\mu M)(s, y)ds + \int_0^y (\mu N)(0, t)dt$, for all $(x, y) \in \mathbb{R}^2$

Question 13 (JAM 2025 • Q29 • ID: JAM_2025_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 Let $\varphi : [0, \infty) \rightarrow \mathbb{R}$ be the continuous function satisfying

$$\varphi(x) = \left(\int_0^x \varphi(t) dt \right) + \sin x, \text{ for all } x \in [0, \infty).$$

Then, the value of $\lim_{x \rightarrow \pi/2} (2\varphi(x) - e^x)$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Question 14 (JAM 2025 • Q14 • ID: JAM_2025_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} = (y - 1)(y - 3),$$

satisfying $\varphi(0) = 2$. Then, which one of the following is TRUE?

(A) $\lim_{x \rightarrow \infty} \varphi(x) = 0$

(B) $\lim_{x \rightarrow \ln \sqrt{2}} \varphi(x) = 1$

(C) $\lim_{x \rightarrow -\infty} \varphi(x) = 3$

(D) $\lim_{x \rightarrow \ln \frac{1}{\sqrt{2}}} \varphi(x) = 6$

Question 15 (JAM 2025 • Q12 • ID: JAM_2025_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $\varphi : (0, \infty) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$x \frac{dy}{dx} = (\ln y - \ln x)y,$$

satisfying $\varphi(1) = e^2$. Then, the value of $\varphi(2)$ is equal to

(A) e^2

(B) $2e^3$

(C) $3e^2$

(D) $6e^3$

Question 16 (JAM 2025 • Q11 • ID: JAM_2025_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 Let $\mathcal{C}$ denote the family of curves described by $yx^2 = \lambda$, for $\lambda \in (0, \infty)$ and lying in the first quadrant of the xy plane. Let $\mathcal{O}$ denote the family of orthogonal trajectories of $\mathcal{C}$.

Which one of the following curves is a member of $\mathcal{O}$, and passes through the point $(2, 1)$?

- (A) $y = \frac{x^2}{4}, x > 0, y > 0$
- (B) $x^2 - 2y^2 = 2, x > 0, y > 0$
- (C) $x - y = 1, x > 0, y > 0$
- (D) $2x - y^2 = 3, x > 0, y > 0$

Question 17 (JAM 2025 • Q5 • ID: JAM_2025_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Which one of the following is the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x} ?$$

- (A) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$
- (B) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$
- (C) $\alpha_1 e^{-4x} + \alpha_2 e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$
- (D) $\alpha_1 x e^{-4x} + \alpha_2 x^2 e^{-4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

Question 18 (JAM 2025 • Q2 • ID: JAM_2025_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 For which one of the following choices of $N(x, y)$, is the equation

$$(e^x \sin y - 2y \sin x) dx + N(x, y) dy = 0$$

an exact differential equation?

- (A) $N(x, y) = e^x \sin y + 2 \cos x$
- (B) $N(x, y) = e^x \cos y + 2 \cos x$
- (C) $N(x, y) = e^x \cos y + 2 \sin x$
- (D) $N(x, y) = e^x \sin y + 2 \sin x$

Question 19 (JAM 2024 • Q51 • ID: JAM_2024_Q51)

NAT

+2 M

Neg: Nil (0)

Section C: Q.51 – Q.60 Carry TWO marks each.Q.51 For $\alpha \in (-2\pi, 0)$, consider the differential equation

$$x^2 \frac{d^2 y}{dx^2} + \alpha x \frac{dy}{dx} + y = 0 \quad \text{for } x > 0.$$

Let D be the set of all $\alpha \in (-2\pi, 0)$ for which all corresponding real solutions to the above differential equation approach zero as $x \rightarrow 0^+$. Then, the number of elements in $D \cap \mathbb{Z}$ equals _____

Question 20 (JAM 2024 • Q43 • ID: JAM_2024_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 For $\alpha > 0$, let $y_\alpha(x)$ be the solution to the differential equation

$$2 \frac{d^2 y}{dx^2} - \frac{dy}{dx} - y = 0$$

satisfying the conditions

$$y(0) = 1, \quad y'(0) = \alpha.$$

Then, the smallest value of α for which $y_\alpha(x)$ has no critical points in $\mathbb{R}$ equals _____ (rounded off to the nearest integer).

Question 21 (JAM 2024 • Q42 • ID: JAM_2024_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let $y: \mathbb{R} \rightarrow \mathbb{R}$ be the solution to the differential equation

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 1$$

satisfying $y(0) = 0$ and $y'(0) = 1$.Then, $\lim_{x \rightarrow \infty} y(x)$ equals _____ (rounded off to two decimal places).

Question 22 (JAM 2024 • Q26 • ID: JAM_2024_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a solution of the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 2e^x \text{ for } x \in \mathbb{R}.$$

Consider the following statements.

P: If $f(x) > 0$ for all $x \in \mathbb{R}$, then $f'(x) > 0$ for all $x \in \mathbb{R}$.Q: If $f'(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) > 0$ for all $x \in \mathbb{R}$.

Then, which one of the following holds?

- (A) P is true but Q is false
- (B) P is false but Q is true
- (C) Both P and Q are true
- (D) Both P and Q are false

Question 23 (JAM 2024 • Q23 • ID: JAM_2024_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 For $\alpha \in \mathbb{R}$, let $y_\alpha(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 2y = \frac{1}{1+x^2} \quad \text{for } x \in \mathbb{R}$$

satisfying $y(0) = \alpha$. Then, which one of the following is TRUE?

- (A) $\lim_{x \rightarrow \infty} y_\alpha(x) = 0$ for every $\alpha \in \mathbb{R}$
- (B) $\lim_{x \rightarrow \infty} y_\alpha(x) = 1$ for every $\alpha \in \mathbb{R}$
- (C) There exists an $\alpha \in \mathbb{R}$ such that $\lim_{x \rightarrow \infty} y_\alpha(x)$ exists but its value is different from 0 and 1
- (D) There is an $\alpha \in \mathbb{R}$ for which $\lim_{x \rightarrow \infty} y_\alpha(x)$ does not exist

Question 24 (JAM 2024 • Q18 • ID: JAM_2024_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 For the differential equation

$$y(8x - 9y)dx + 2x(x - 3y)dy = 0,$$

which one of the following statements is TRUE?

- (A) The differential equation is not exact and has x^2 as an integrating factor
- (B) The differential equation is exact and homogeneous
- (C) The differential equation is not exact and does not have x^2 as an integrating factor
- (D) The differential equation is not homogeneous and has x^2 as an integrating factor

Question 25 (JAM 2024 • Q10 • ID: JAM_2024_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Which one of the following is the solution of the differential equation

$$\frac{d^2 y}{dx^2} + y = g(x) \text{ for } x \in \mathbb{R},$$

satisfying the conditions $y(0) = 0$, $y'(0) = 1$?

(A) $y(x) = \sin x - \int_0^x \sin(x-t) g(t) dt$

(B) $y(x) = \sin x + \int_0^x \sin(x-t) g(t) dt$

(C) $y(x) = \sin x - \int_0^x \cos(x-t) g(t) dt$

(D) $y(x) = \sin x + \int_0^x \cos(x-t) g(t) dt$

Question 26 (JAM 2024 • Q4 • ID: JAM_2024_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let $\mathcal{F}$ be the family of curves given by

$$x^2 + 2hxy + y^2 = 1, \quad -1 < h < 1.$$

Then, the differential equation for the family of orthogonal trajectories to $\mathcal{F}$ is

(A) $(x^2y - y^3 + y)\frac{dy}{dx} - (xy^2 - x^3 + x) = 0$

(B) $(x^2y - y^3 + y)\frac{dy}{dx} + (xy^2 - x^3 + x) = 0$

(C) $(x^2y + y^3 + y)\frac{dy}{dx} - (xy^2 + x^3 + x) = 0$

(D) $(x^2y + y^3 + y)\frac{dy}{dx} + (xy^2 + x^3 + x) = 0$

Question 27 (JAM 2024 • Q3 • ID: JAM_2024_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = 1 + y \sec x \quad \text{for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

that satisfies $y(0) = 0$. Then, the value of $y\left(\frac{\pi}{6}\right)$ equals

- (A) $\sqrt{3} \log\left(\frac{3}{2}\right)$
- (B) $\left(\frac{\sqrt{3}}{2}\right) \log\left(\frac{3}{2}\right)$
- (C) $\left(\frac{\sqrt{3}}{2}\right) \log 3$
- (D) $\sqrt{3} \log 3$

Question 28 (JAM 2024 • Q1 • ID: JAM_2024_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Let $y_c: \mathbb{R} \rightarrow (0, \infty)$ be the solution of the Bernoulli's equation

$$\frac{dy}{dx} - y + y^3 = 0, \quad y(0) = c > 0.$$

Then, for every $c > 0$, which one of the following is true?

- (A) $\lim_{x \rightarrow \infty} y_c(x) = 0$
- (B) $\lim_{x \rightarrow \infty} y_c(x) = 1$
- (C) $\lim_{x \rightarrow \infty} y_c(x) = e$
- (D) $\lim_{x \rightarrow \infty} y_c(x)$ does not exist

Question 29 (JAM 2023 • Q59 • ID: JAM_2023_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let S be the set of all real numbers α such that the solution y of the initial value problem

$$\begin{aligned} \frac{dy}{dx} &= y(2 - y), \\ y(0) &= \alpha, \end{aligned}$$

exists on $[0, \infty)$. Then the minimum of the set S is equal to _____.
(rounded off to two decimal places)

Question 30 (JAM 2023 • Q53 • ID: JAM_2023_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 Let $y : (1, \infty) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$y'' - \frac{2y}{(1-x)^2} = 0$$

satisfying $y(2) = 1$ and $\lim_{x \rightarrow \infty} y(x) = 0$. Then $y(3)$ is equal to _____.
(rounded off to two decimal places)

Question 31 (JAM 2023 • Q43 • ID: JAM_2023_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 If y is the solution of

$$y'' - 2y' + y = e^x,$$
$$y(0) = 0, \quad y'(0) = -1/2,$$

then $y(1)$ is equal to _____. (rounded off to two decimal places)

Question 32 (JAM 2023 • Q33 • ID: JAM_2023_Q33)

MSQ

+2 M

Neg: Nil (0)

Q. 33 Let $y : (\sqrt{2/3}, \infty) \rightarrow \mathbb{R}$ be the solution of

$$(2x - y)y' + (2y - x) = 0,$$

$$y(1) = 3.$$

Then

- (A) $y(3) = 1$
- (B) $y(2) = 4 + \sqrt{10}$
- (C) y' is bounded on $(\sqrt{2/3}, 1)$
- (D) y' is bounded on $(1, \infty)$

Question 33 (JAM 2023 • Q22 • ID: JAM_2023_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 Consider the family of curves $x^2 + y^2 = 2x + 4y + k$ with a real parameter $k > -5$. Then the orthogonal trajectory to this family of curves passing through $(2, 3)$ also passes through

- (A) $(3, 4)$
- (B) $(-1, 1)$
- (C) $(1, 0)$
- (D) $(3, 5)$

Question 34 (JAM 2023 • Q4 • ID: JAM_2023_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Consider the initial value problem

$$\begin{aligned}\frac{dy}{dx} + \alpha y &= 0, \\ y(0) &= 1,\end{aligned}$$

where $\alpha \in \mathbb{R}$. Then

- (A) there is an α such that $y(1) = 0$
- (B) there is a unique α such that $\lim_{x \rightarrow \infty} y(x) = 0$
- (C) there is NO α such that $y(2) = 1$
- (D) there is a unique α such that $y(1) = 2$

Question 35 (JAM 2022 • Q50 • ID: JAM_2022_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50

Consider the family $\mathcal{F}_1$ of curves lying in the region

$$\{(x, y) \in \mathbb{R}^2 : y > 0 \text{ and } 0 < x < \pi\}$$

and given by

$$y = \frac{c(1 - \cos x)}{\sin x}, \text{ where } c \text{ is a positive real number.}$$

Let $\mathcal{F}_2$ be the family of orthogonal trajectories to $\mathcal{F}_1$. Consider the curve $\mathcal{C}$ belonging to the family $\mathcal{F}_2$ passing through the point $\left(\frac{\pi}{3}, 1\right)$. If a is a real number such that $\left(\frac{\pi}{4}, a\right)$ lies on $\mathcal{C}$, then the value of a^4 is equal to _____.
(Rounded off to two decimal places)

Question 36 (JAM 2022 • Q49 • ID: JAM_2022_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49

Given that $y(x)$ is a solution of the differential equation

$$x^2 y'' + x y' - 4y = x^2$$

on the interval $(0, \infty)$ such that $\lim_{x \rightarrow 0^+} y(x)$ exists and $y(1) = 1$. The value of $y'(1)$ is equal to _____. (Rounded off to two decimal places)

Question 37 (JAM 2022 • Q47 • ID: JAM_2022_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47

Let $y(x)$ be the solution of the differential equation

$$x y^2 y' + y^3 = \frac{\sin x}{x} \quad \text{for } x > 0,$$

satisfying $y\left(\frac{\pi}{2}\right) = 0$.

Then the value of $y\left(\frac{5\pi}{2}\right)$ is equal to _____. (Rounded off to two decimal places)

Question 38 (JAM 2022 • Q43 • ID: JAM_2022_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43

Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 3x^2 y = x^2, \quad \text{for } x \in \mathbb{R},$$

satisfying the initial condition $y(0) = 4$.

Then $\lim_{x \rightarrow \infty} y(x)$ is equal to _____. (Rounded off to two decimal places)

Question 39 (JAM 2022 • Q35 • ID: JAM_2022_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35

A real-valued function $y(x)$ defined on $\mathbb{R}$ is said to be **periodic** if there exists a real number $T > 0$ such that $y(x + T) = y(x)$ for all $x \in \mathbb{R}$.

Consider the differential equation

$$\frac{d^2y}{dx^2} + 4y = \sin(ax), \quad x \in \mathbb{R}, \quad (*)$$

where $a \in \mathbb{R}$ is a constant.

Then which of the following is/are true?

- | | |
|-----|--|
| (A) | All solutions of (*) are periodic for every choice of a . |
| (B) | All solutions of (*) are periodic for every choice of $a \in \mathbb{R} - \{-2, 2\}$. |
| (C) | All solutions of (*) are periodic for every choice of $a \in \mathbb{Q} - \{-2, 2\}$. |
| (D) | If $a \in \mathbb{R} - \mathbb{Q}$, then there is a unique periodic solution of (*). |

Question 40 (JAM 2022 • Q31 • ID: JAM_2022_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31	Let $(-c, c)$ be the largest open interval in $\mathbb{R}$ (where c is either a positive real number or $c = \infty$) on which the solution $y(x)$ of the differential equation $\frac{dy}{dx} = x^2 + y^2 + 1 \quad \text{with initial condition} \quad y(0) = 0$ exists and is unique. Then which of the following is/are true?
(A)	$y(x)$ is an odd function on $(-c, c)$.
(B)	$y(x)$ is an even function on $(-c, c)$.
(C)	$(y(x))^2$ has a local minimum at 0.
(D)	$(y(x))^2$ has a local maximum at 0.

Question 41 (JAM 2022 • Q25 • ID: JAM_2022_Q25)

MCQ

+2 M

Neg: -0.67

Q.25

For some real number c with $0 < c < 1$, let $\phi: (1 - c, 1 + c) \rightarrow (0, \infty)$ be a differentiable function such that $\phi(1) = 1$ and $y = \phi(x)$ is a solution of the differential equation

$$(x^2 + y^2)dx - 4xy dy = 0.$$

Then which one of the following is true?

(A)

$$\left(3(\phi(x))^2 + x^2\right)^2 = 4x.$$

(B)

$$\left(3(\phi(x))^2 - x^2\right)^2 = 4x.$$

(C)

$$\left(3(\phi(x))^2 + x^2\right)^2 = 4\phi(x).$$

(D)

$$\left(3(\phi(x))^2 - x^2\right)^2 = 4\phi(x).$$

Question 42 (JAM 2022 • Q23 • ID: JAM_2022_Q23)

MCQ

+2 M

Neg: -0.67

Q.23	Consider the differential equation $y'' + ay' + y = \sin x \quad \text{for } x \in \mathbb{R}. \quad (**)$ Then which one of the following is true?
(A)	If $a = 0$, then all the solutions of $(**)$ are unbounded over $\mathbb{R}$.
(B)	If $a = 1$, then all the solutions of $(**)$ are unbounded over $(0, \infty)$.
(C)	If $a = 1$, then all the solutions of $(**)$ tend to zero as $x \rightarrow \infty$.
(D)	If $a = 2$, then all the solutions of $(**)$ are bounded over $(-\infty, 0)$.

Question 43 (JAM 2022 • Q21 • ID: JAM_2022_Q21)

MCQ

+2 M

Neg: -0.67

Q.21	On the open interval $(-c, c)$, where c is a positive real number, $y(x)$ is an infinitely differentiable solution of the differential equation $\frac{dy}{dx} = y^2 - 1 + \cos x,$ with the initial condition $y(0) = 0$. Then which one of the following is correct?
(A)	$y(x)$ has a local maximum at the origin.
(B)	$y(x)$ has a local minimum at the origin.
(C)	$y(x)$ is strictly increasing on the open interval $(-\delta, \delta)$ for some positive real number δ .
(D)	$y(x)$ is strictly decreasing on the open interval $(-\delta, \delta)$ for some positive real number δ .

Question 44 (JAM 2022 • Q13 • ID: JAM_2022_Q13)

MCQ

+2 M

Neg: -0.67

Q.13

Let $u: \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously differentiable function such that $u(0) > 0$ and $u'(0) > 0$. Suppose u satisfies

$$u''(x) = \frac{u(x)}{1+x^2} \text{ for all } x \in \mathbb{R}.$$

Consider the following two statements:

I. The function uu' is monotonically increasing on $[0, \infty)$.

II. The function u is monotonically increasing on $[0, \infty)$.

Then which one of the following is correct?

(A) Both I and II are false.

(B) Both I and II are true.

(C) I is false, but II is true.

(D) I is true, but II is false.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+2	-1.0 to -1.0	16	MCQ	+2	B	31	NAT	+1	-0.01 to 0.01
2	NAT	+1	8.0 to 8.0	17	MCQ	+1	A	32	MSQ	+2	B, D
3	MSQ	+2	B;D	18	MCQ	+1	B	33	MCQ	+2	A
4	MCQ	+2	B	19	NAT	+2	6 to 6	34	MCQ	+1	D
5	MCQ	+2	D	20	NAT	+1	1.0 to 1.0	35	NAT	+1	2.00 to 2.00
6	MCQ	+2	C	21	NAT	+1	0.19 to 0.21	36	NAT	+1	2.24 to 2.26
7	MCQ	+1	D	22	MCQ	+2	B	37	NAT	+1	0 to 0
8	NAT	+2	3.98 to 4.02	23	MCQ	+2	A	38	NAT	+1	0.32 to 0.34
9	NAT	+2	0.98 to 1.02	24	MCQ	+2	A	39	MSQ	+2	C,D
10	NAT	+1	1.39 to 1.43	25	MCQ	+1	B	40	MSQ	+2	A,C
11	MSQ	+2	A;B;D	26	MCQ	+1	A	41	MCQ	+2	B
12	MSQ	+2	A;B;C	27	MCQ	+1	A	42	MCQ	+2	A
13	MCQ	+2	A	28	MCQ	+1	B	43	MCQ	+2	D
14	MCQ	+2	C	29	NAT	+2	-0.01 to 0.01	44	MCQ	+2	B
15	MCQ	+2	B	30	NAT	+2	0.49 to 0.51				

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.